

İTÜ

Security in Financial Information Systems

Financial Markets and Currency

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Motivation

Thodex'in kurucusu hakkında kırmızı bülten talebi



Kripto para borsası Thodex'in kurucusu ve yöneticisi Faruk Fatih Özer hakkında kırmızı bülten çıkarıldı. Bakan Soylu da Arnavutluk İçişleri Bakanı ile bir görüşme yaptı. Zanlının Türkiye'ye iadesi isteniyor. Devam eden soruşturma kapsamında 62 kişi gözaltına alındı, 16 kişi de aranıyor.

<https://www.ntv.com.tr/turkiye/son-dakika-haberi-istanbul-l-dahil-8-ilde-thodex-operationsu-AM6kxw008kUW000y6>

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Outline

- Financial Markets
- Money and Currency

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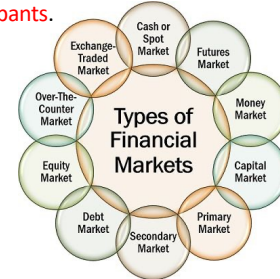
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Financial Markets

Financial markets are mechanisms and conventions that exist for transfer of funds and their counterparts between the various participants.



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Financial Markets

All ultimate lenders and borrowers and all financial intermediaries are participants in financial markets.

- Household,
- Corporate,
- Government,
- Foreign sectors
- ...



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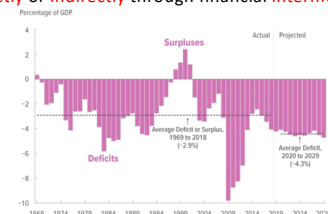
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Financial Markets

Financial markets constitute the mechanism that links surplus and deficit units -> providing means for surplus units to finance deficit units either directly or indirectly through financial intermediaries.



Financial markets provide surplus and deficit units with additional options.

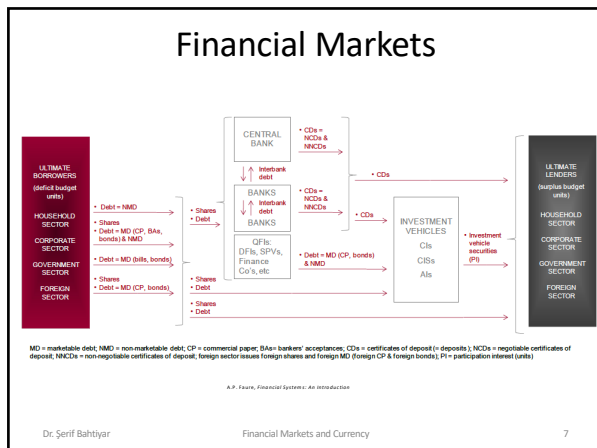
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Financial Markets



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Financial Markets

Types of Financial Markets

- Money market.
- Bond market.
- Share market.
- Foreign exchange market.
- Derivatives markets.
- Financial market participants and short selling.
- Clearing and settlement.

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Financial Markets

There are two broad categories of securities:

- Debt and deposits
 - Securities issued by the ultimate borrowers
 - Marketable debt
 - Non-marketable debt
 - Categorized into the money market and the bond market
 - Deposit securities of the banks
 - Negotiable certificates of deposit
 - Non-negotiable certificates of deposit
 - Notes and coins (central bank)
 - Central bank securities
- Shares

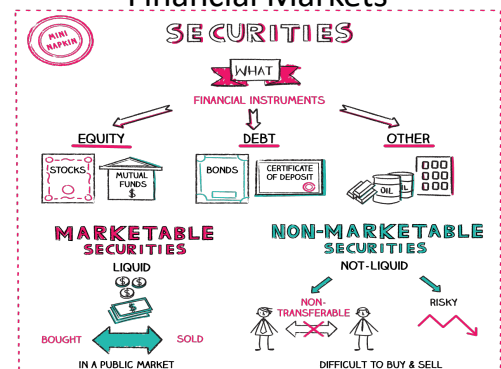
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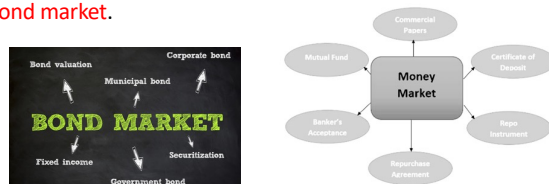
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Financial Markets

Debt (deposit) market is categorized into money market and bond market.



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Financial Markets

Money market also contains

- Money creation by banks
- Interbank markets
 - Bank-to-bank interbank market
 - Bank-to-central bank interbank market
 - Central bank-to-bank interbank market



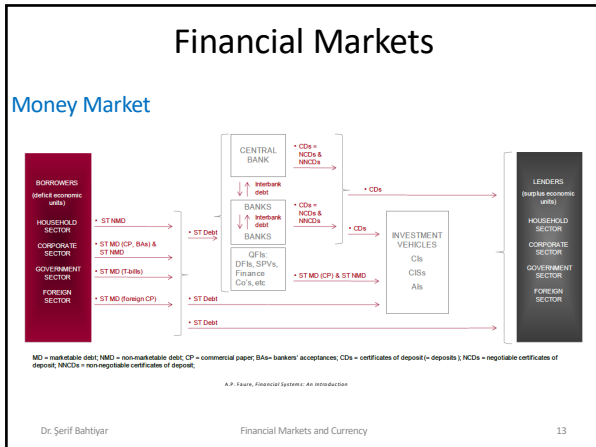
Banks are required to settle amounts owed to one another at the end of each business day. This occurs over accounts held with central bank.

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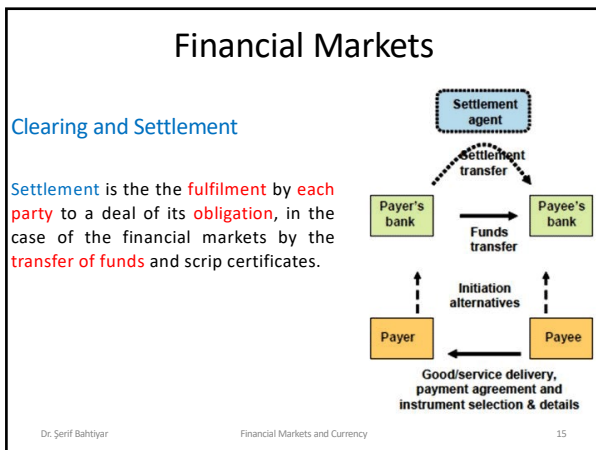
Clearing and Settlement

Clearing is the **process** of determining **accountability** for the exchange of funds and financial assets among the parties to a financial transaction.

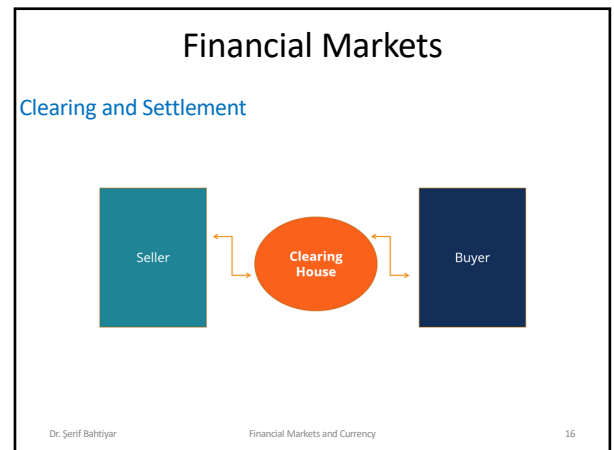
- Involves **reporting** details of the trade to counterparties, clients, exchanges, regulatory bodies, and others.
- Involves **matching** or **comparing** the details of the trade with **clients** and **counterparties** to **make certain all the parties agree**.

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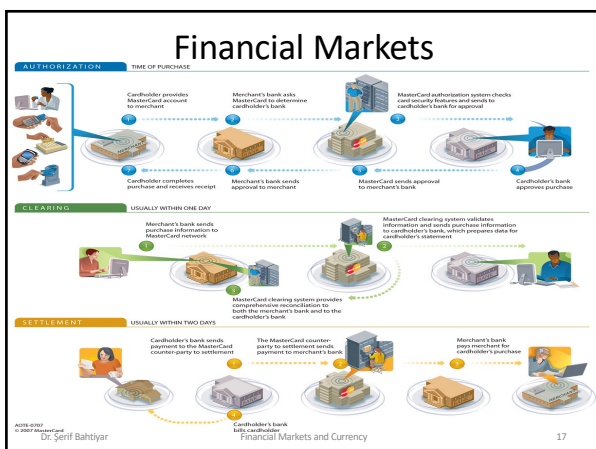
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Financial Markets

Why do we explain financial markets and some related concepts?

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Money

What is money?

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Money and Currency

Money

- **Symbol** devised for the purpose of **exchanges**. (*Plato*)
- **Guarantee** that we may have what **we want** in the **future**. (*Aristotle*)
- **Any item** or **verifiable record** that is generally **accepted as payment** for **goods** and **services** and **repayment** of **debts**, such as taxes, in a particular country or socio-economic context. (*Wikipedia*)



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Currency

Currency is **money** in **any form** when in **use** or **circulation** as a **medium of exchange**, especially circulating **banknotes** and **coins**. (*Wikipedia*)



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Money and Currency

Properties of Currency

- **Medium of exchange**
- **Unit of account**
- **Portable**



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Properties of Currency-2

- **Durable**
- **Divisible**
- **Interchangeable**



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Money and Currency

Properties of Money

All properties of currency
+
Store of Value



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Fiat Currency

- A **fiat currency** is a **national currency** that is **not pegged** to the price of a commodity such as **gold** or **silver**.
- Depends on **trusted authority**, such as governments.



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Money and Currency

Fiat Currency

- Represented by**
 - Central Bank Notes
 - Central Bank Reserves
 - Commercial Bank Deposits
- Relies on System of Ledgers**
- Significant Network Effects**
 - Accepted for taxes
 - Accepted by economy



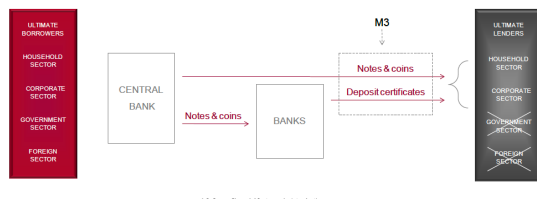
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A.P. Evans, Financial Systems: An Introduction

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Digital Currency

Digital currency or **digital money** is **distinct** from **physical** that exhibits properties **similar to physical currencies**, but **allows** for **instantaneous transactions** and **borderless** transfer-of-ownership. (Wikipedia)

- B-Money
- E-Gold
- ...



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Cryptocurrency

- Cryptocurrency** is a **peer-to-peer** version of electronic cash.
- Does not need a **trusted-third party** (financial institution) for payments.
 - eCash
 - Bitcoin
 - ...
- Based on **cryptography**



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Cryptocurrency

- Cryptocurrency** allows online payments to be sent **directly** from one party to another **without going through a financial institution**.



- The **loss of trust in fiat currency** system has brought attention to **cryptocurrency**.

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Cryptocurrency

Common concerns against cryptocurrency

- Lack of a **central issuing authority** (e.g. a central bank)
- **Fixed** supply and deflationary nature by design
- **Unstable** price to function as a currency
- **Security** issues



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Summary

- Financial markets
- Types of financial markets
- Money market
- Clearing and settlement
- Money
- Currency
- Properties of currency
- Fiat currency
- Digital currency
- Cryptocurrency

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Questions?

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